A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 22

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 38.0 / 0.002 **1** = 19 000 (19 000 times / 1.9 x 10^4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • A bare wrong answer with no working. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	1 mark for the point below. **1** Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biomagnification** → concentration rising at each step along a food chain • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	1
01.3	**1 mark each** 1 mark for each of the 4 points below. **1 Persistence:** the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain **1 Liposolubility:** the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine **1** Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level **1** Herons have long lives so accumulate over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Persistence** → does not break down / stays in the environment for a long time • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
01.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification **1** They persist for decades so the damage continued long after use **1** Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify **1** Benefit of preventing malaria deaths outweighs the risk (risk:benefit) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ground surface:** pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals – trapped predators eat others – preservative attracts / repels – escapes **2 Leaf litter:** Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light – some die or stop moving – large ones cannot pass the mesh **2 Branches:** beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour – some species not dislodged / fly away – high branches unreachable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pitfall trap** → a sunken container that catches ground invertebrates • **Tullgren** → a funnel that drives soil invertebrates down into a collecting jar	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught **1** Brightness / wavelength of the lamp affects attractiveness so catches vary between traps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The area of ground covered by the suction pipe is known so invertebrates per m^2 can be calculated **1** A pooter is only a handling / collection device for individuals already found, not a sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
03.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) **1** They work best in sheltered / calm water such as the lagoon **1 Limitation:** they do not work in open water where currents or waves are strong – oil can flow under the skirt in a strong current **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Skimmers:** rotating (metal) discs / belts pick up floating oil which is scraped off and stored **1** Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed **1** Collected oil / absorbents are incinerated or sent to landfill (or oil recycled) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface **1** They do not reduce the amount of oil – the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) **1** Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Most sulfur in coal is present as (solid) iron pyrites (FeS2) **1** Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Flue gases are passed through a bed of crushed calcium carbonate (limestone) **1** Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) **1** If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide **1** The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	2
05.4	**1 mark each** 1 mark for each of the 3 points below. **1 Fluidised bed:** increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NOx (production prevented) **1 Catalytic converter:** platinum / palladium catalyst chemically reduces NOx back to nitrogen and oxygen gases in the exhaust **1** Urea sprayed into the hot flue / exhaust gases reacts with NOx, converting it to nitrogen, carbon dioxide and water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Orientation:** south-facing windows (northern hemisphere) maximise passive solar gains – rooms needing warmth on the sunny side – fewer / smaller windows on the north side where losses are greatest – solar screens / brise soleil prevent summer overheating **2** Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost **2 Earth sheltering:** the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy **1** Limecrete has about half the embodied energy of concrete – rammed earth, straw bales and lime mortar are lower still **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **embodied energy** → all the energy used to make and transport a product	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area **1** Heat transfers from the outgoing to the incoming air so fresh air enters already warmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations **1** It bioaccumulated in shellfish and could pass to humans eating them **1** It was assumed dilution would make it harmless (illustrates the need for the precautionary principle) – now prohibited and replaced by copper **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	3
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river **1** In the river it is oxidised to insoluble iron oxide (orange sediment) – the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms **1 Control:** drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river – the solid iron is removed periodically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
07.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lead-acid car batteries (recycled) **1** Lead flashing on roofs / windows **1** Radiation shielding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.5	1 mark for the point below. **1** They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 + 5 = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • A bare wrong answer with no working. • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	1
08.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 UV:** short wavelength – mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium – converted to chemical energy in these reactions – little reaches the surface **2 Visible light:** transmitted through the atmosphere easily (some reflected by clouds / dust) – absorbed by the Earth's surface and converted to heat – some absorbed by chlorophyll and converted to chemical energy in photosynthesis **2 IR:** emitted by the warm surface (long-wave) – absorbed by greenhouse gases (CO2, water vapour, methane) and converted to heat – the warm atmosphere re-emits IR to the surface and reduces conduction losses – some escapes to space **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down **1** The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy – only the net loss (70%) balances the absorbed solar energy at equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
08.4	1 mark for the point below. **1** The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Trees absorb CO2 by photosynthesis and store carbon in wood as they grow **1** Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir **1** Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 4 points below. **1 Pre-combustion:** the fuel is modified before use so no CO2 is released when burnt - eg gasification of coal to hydrogen and CO2 (CO2 stored, hydrogen distributed) – oxy-fuel combustion in pure oxygen giving only CO2 and water vapour which is condensed out **1 Advantage:** capture at a large central facility – suits small dispersed sources (vehicles) that cannot capture CO2 **1 Post-combustion:** CO2 removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene **1 Disadvantage:** only about 20% of flue gas is CO2 so separation is costly / energy-intensive – none fully developed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	4
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Injected into depleted oil / gas fields or saline aquifers **1** Injected into oil reservoirs to provide pressure for secondary oil recovery **1** Reaction with fractured basalt to form solid carbonate minerals **1 Concern:** leakage back to the atmosphere / monitoring over centuries – limited suitable geology – cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Properties of renewables:** intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • **Wind:** large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments – intermittent, needs storage and grid – land-use conflicts • **Solar:** PV costs falling rapidly, roof space, but low winter insolation in the UK – CSP unsuitable – heat pumps and photothermal for heating • **HEP:** most large sites used – low-head, helical and micro-hydro extend use – pumped storage vital • **Tidal:** large range (Severn, Mersey), predictable – barrages costly with high impact, lagoons and in-stream turbines lower impact • **Wave:** long Atlantic fetch, but technology immature • **Biofuels:** wastes, anaerobic digestion, energy crops vs food / habitat • **Geothermal:** limited to low-temperature schemes / heat pumps • **Systems issues:** storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • **Judgement:** renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder – transition needs investment and lifestyle change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any accurate, relevant material that is not in the indicative content. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **insolation** → incoming solar radiation / energy arriving from the Sun • **HEP** → electricity generated from falling water **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (8 marks)	
	What renewables can contribute depends on their properties - intermittency, predictability, energy density, storage, applicability, location, impacts, technological maturity and cost - set against the UK's resources. Wind is the largest: shallow seas give a huge offshore resource, costs have fallen and direct-drive and vertical-axis designs keep improving, but it is intermittent, needs storage and grid connections and provokes land-use conflicts onshore. Solar PV costs are falling rapidly and roof space is plentiful, but winter insolation is low and concentrating solar is unsuitable; photothermal panels and heat pumps are better for heating. Most large HEP sites are used, though low-head, helical and micro-hydro extend it and pumped storage is vital for balancing. The Severn and Mersey have large, predictable tidal ranges, but barrages are costly and damaging while lagoons and in-stream turbines have lower impact; the long Atlantic fetch gives wave energy that the technology cannot yet harness reliably. Biofuels from wastes and anaerobic digestion are useful, but energy crops compete with food and habitat, and geothermal is limited to low-temperature schemes.	
	The system matters as much as the sources: storage in pumped HEP, batteries, hydrogen and power-to-gas, vehicle-to-grid, grid management, a diverse mix that smooths supply, and conservation to reduce demand. With these, renewables could supply most of the UK's electricity. Heat and transport are harder, because they need high temperatures or liquid fuels, so electrification and hydrogen would be required. The potential is large, but realising it needs sustained investment and changes in how energy is used.	
	Weaker answer - Level 1 (3 marks)	
	The UK could use lots of renewable energy. It is windy so wind turbines could be built, especially offshore. Solar panels can be put on roofs but it is not very sunny in winter. Tidal power could be used in the Severn estuary because it has big tides. Hydroelectric power can be used in Scotland. Renewables are intermittent so batteries would be needed. Renewables could provide a lot of the UK's energy in the future.	
	What makes the difference	
	• Level 1 mentions the main resources with one fact each; Level 3 assesses each against its properties, constraints and the state of the technology.	
	• Level 3 treats storage, grid and demand as a system and distinguishes electricity from heat and transport - essential for judging 'future energy supplies'.	

Do not write outside the box

Question	Marking guidance	Marks
	• The conclusion in Level 3 is qualified (most electricity; heat and transport harder; needs investment); Level 1's is unqualified.	
10.2	1 mark for the point below. **1** Eight times (energy proportional to velocity cubed) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

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